Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*High-reliability SWIR image sensor for automotive harsh-condition environment perception and autonomous driving recognition integration technology*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical, Materials		*Electrical/Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile/Rail Vehicles	*Safety Improvement Technologies*	*Semiconductor Devices and Systems*	*SoC*
Concept of Proposed Technology	The proposed technology develops SWIR-based image sensor technology to overcome the limitations of conventional visible-light cameras in automotive CCTV and autonomous driving systems. The SWIR band (900–1700 nm) offers superior transmission and material identification capabilities compared with visible light even in fog, smoke, dust, rain and snow, backlighting, and nighttime environments, enabling stable visual perception in harsh driving conditions. The proposed technology integrates the following elements: • Integration and packaging design of high-resolution InGaAs photodetectors with a CMOS ROIC • Robust low-noise analog frontend and drive module design suitable for CCTV/vehicle environments • AI-based object recognition and hazard detection algorithms optimized for SWIR image characteristics • Development of a high-reliability, low-power SWIR image sensor prototype considering			

	CCTV/vehicle standards Through these developments, a next-generation non-visible-light sensing platform can be realized that complements or replaces existing automotive CCTV, forward/rear cameras, surround view systems, and autonomous driving recognition cameras.
Rationale for Joint R&D Project	This international joint research and development project is pursued based on a collaborative structure with clearly defined roles among overseas research institutions holding world-class foundational technology in the SWIR image sensor field, domestic universities, and industry partners. • **Korea Partner(s):** responsible for developing foundational technologies for ROIC (Readout Integrated Circuit) design and SWIR camera modules (128×128 / 320×320 / 640×640 resolutions) A company, which can validate the developed SWIR sensors and camera modules in automotive CCTV and autonomous driving recognition systems, and will lead technology enhancement and commercialization strategy aligned with real market needs. • **Fraunhofer Institute**: A world-leading research organization in the SWIR image sensor field. It was the first in the world to implement a 96×96 pixel SWIR image sensor array, a result that demonstrates both technical leadership and reliability in the field. Fraunhofer possesses unique foundational technologies and extensive research know-how in high-performance SWIR photodetectors and pixel array design and will be responsible for advancing the core photodetector array technology and ensuring reliability in this project. This collaboration enables achieving world-class SWIR sensor technology that would be difficult for a single institution to accomplish, while also enabling real-world industrial application in the automotive and autonomous driving fields.
Key Words	1. Automotive SWIR image sensor 2. High-resolution SWIR pixel array 3. Low-noise, low-power ROIC 4. Harsh-condition environment perception 5. CCTV/autonomous driving ADAS camera system

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	1. Securing core SWIR image sensor technology for automotive CCTV and autonomous driving applications 2. Development of SWIR-based intelligent recognition: classification and detection algorithms 3. Fusion sensor for SWIR–visible light–LIDAR: Integration of InGaAs photodetectors with CMOS ROIC and sensor module implementation 4. Validation of recognition stability in real-world automotive test environments 5. Ensuring scalability to a vehicle sensor platform: Verification of SWIR sensor system			

	prototypes considering industrial and mobility applications
R&D Target Specifications	• Recognizable environment: fog, smoke, rain, snow, nighttime driving environment • Operating wavelength band: 900 ~ 1700 nm • Effective resolution: at least 96 × 96 • Object recognition distance: at least 1.5× greater than visible light • Vehicle real-time processing latency: ≤ 50 ms
R&D Plan	**Phase 1 (Year 1):** Core technology design and feasibility study ✓ Analysis of SWIR photodetector characteristics and requirements definition ✓ Low-noise, low-power ROIC architecture study ✓ Developments of pixel/column-level verification method ✓ Feasibility study of acquisition of SWIR image data and image processing algorithms **Phase 2 (Year 2):** Sensor integration and prototype fabrication ✓ InGaAs–CMOS ROIC integration structure design ✓ Fabrication of test chips and evaluation of electrical and optical characteristics ✓ Sensor module (packaging) implementation ✓ AI-based SWIR image pre-processing and recognition algorithm enhancement **Phase 3 (Year 3):** System verification and demonstration ✓ Vehicle-mounted SWIR sensor module fabrication ✓ Development of AI algorithms for object, pedestrian, and obstacle recognition

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Fraunhofer Institute:** • Based on the world's first 96×96 pixel SWIR image sensor array technology • Design and optimization of high-sensitivity SWIR photodetector pixel structures • Design/analysis and advancement of pixel uniformity, sensitivity, and noise characteristics • Assessment of scalability to next-generation high-resolution SWIR pixel arrays • Will provide the core foundational technology related to photodetector arrays and assume a technical leadership role in the project •
Expected R&R of Korean SME Partner	**Korean partner(s):** • Design of low-noise, low-power ROIC architectures for SWIR image sensors • Development of analog frontend, column-parallel readout circuitry, and high-speed, high-reliability signal processing circuits • Co-design optimization of SWIR pixel arrays and ROIC • Design of SWIR camera modules and system-level performance verification • Development of signal correction and AI-based image processing algorithms optimized for SWIR imaging • Defining system requirements from the perspective of CCTV and vehicle autonomous driving/ADAS cameras • Design and integration of camera systems based on SWIR image sensors • System verification for CCTV and autonomous driving recognition scenarios in vehicles

	• Real-world validation tests and performance evaluation in adverse weather, nighttime, and backlighting environments • Verification of improved recognition performance of SWIR sensors compared with visible-light cameras

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical achievements** • Securing a high-performance SWIR image sensor by combining world-class SWIR pixel array technology and ROIC technology • Establishing vehicle SWIR sensor technology capable of stable recognition in harsh conditions (fog, smoke, rain, snow, nighttime) • Implementing a next-generation intelligent image sensor platform integrating sensor–circuit–AI algorithms • Laying the technological foundation for future expansion to high-resolution, large-area SWIR sensors **Industrial and commercialization achievements** • Securing differentiated near-infrared video solutions targeting the automotive CCTV and autonomous driving/ADAS camera market • Creating a high-value sensor product line that complements existing visible-light cameras and LiDAR • Reducing import dependency and enabling technology self-reliance by domestic SWIR sensor system capabilities • Enabling demonstration–mass production–commercialization linkage centered on ModiDents Vision **Impact** • Strengthening competitiveness in advanced mobility and autonomous driving safety technologies • Enhancing domestic technological standing in SWIR image sensors and expanding global collaboration • Building a foundation for domestic SMEs to enter global markets through international joint research • Enabling potential expansion into security, defense, industrial inspection, and environmental monitoring applications